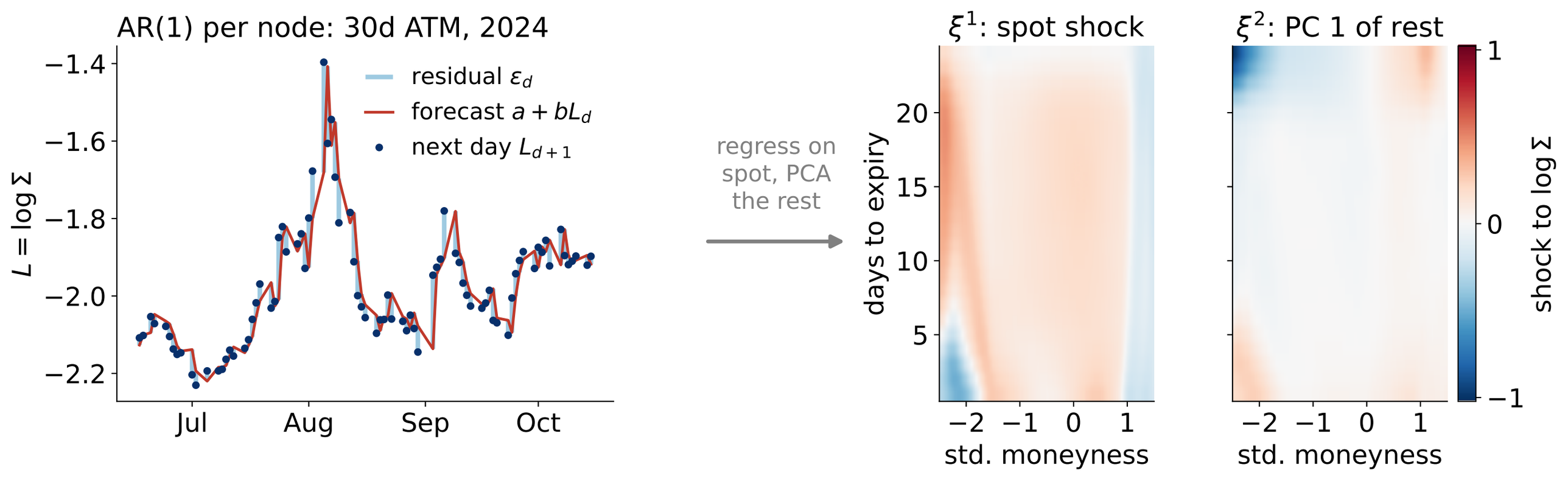

AR(1) per node: 30d ATM, 2024
residual ε_d
forecast $a + bL_d$
next day L_{d+1}
$L = \log \Sigma$
−1.4
−1.6
−1.8
−2.0
−2.2
Jul
Aug
Sep
Oct
regress on spot, PCA the rest
ξ^1: spot shock
ξ^2: PC 1 of rest
days to expiry
20
15
10
5
−2 −1 0 1
std. moneyness
shock to log Σ
1
0
−1